

26 – M6 POS & Offline

Module design for the till: selling, tendering as record, printing trilingual receipts, deducting stock, running a session with blind close, hosting the shop-management screens, and doing all of it for two days without central – then synchronising without ever merging. Also the central side that ingests, re-validates and watches the fleet.

Field	Value
Document	26 – M6 POS & Offline (Stage 2 module design; template of doc 20 §5)
Version	v0.1 – draft for design review; contract freeze targeted end of Stage 2 week 6
Date	9 September 2026
Owning team	The till track (native Android, doc 30 DR-1); the central ingestion side is a thin M6 package in the monolith. M4 and M5 supply shop-mode screen specifications; M3 supplies the engine; the kernel supplies the sync gateway.
Inputs	05 SRS (FR-POS-010–110, FR-PAY-010, 030–050, FR-REP-030); 03 §13; 13 scenarios 2 and 4; 16 answers (H, I-01, I-03, N-03, N-09, O-02, P-05, Q-07); 09 ADR-10, 19, 24, 31, 34, 36; 14 §3, §8; 17 v0.2.1 §9; 18 v0.2; 19 v0.2; 20 v0.3 M6; 21–25 contracts and bundle specs; 30 §2, §4; 31; 32; the multi-MRP study
Depends on	Snapshots from M1 (operators, permissions), M2 (catalogue, batches, images), M3 (prices, rules, ceilings, policies), M5 (lots, count tasks, policies), M4 (expected drops, transfers), M7 (customers, balances); the shared engine; the kernel sync gateway and management agent
Depended on by	M5 (sale deductions from receipt bundles), M7 (account tenders), M4 (GRN bundles), M8 (till sessions, variance, sales), doc 31 (health and version telemetry)

The till is a small, complete system. It has its own database, its own gapless counters, its own copy of the rules, its own audit log and its own outbox. Central never edits what it produces and it never edits what central sends. That symmetry (ADR-10) is why there is no conflict resolution anywhere; every other design choice in this module protects it. The second commitment is speed at the counter: three taps to sell, and nothing that requires reading more than a word.

1. Purpose and scope

In scope (phase 1)	Out of scope
Selling: scan, code lookup, search by name in any script, weighed goods (scale or keyed weight, 20–29 labels), thumbnails; price and rules through the shared engine; multi-MRP policy incl. the picker; control-price cap; negative-stock acknowledgement; sell-through of inactive items (H-02, FR-POS-010–040)	Card, wallet or QR processing; any payment gateway (ADR-19)
Tender recording: cash with rupee rounding and change; card reference; customer account within the offline cap; mixed tenders; vouchers and subsidy instruments later (I-01, FR-PAY-010)	Self-checkout, home delivery, online ordering (H-14)
Receipt as an immutable fiscal document from the till-position series; trilingual raster printing; reprint audited; customer display (H-09, P-05, N-05)	Per-transaction language switch (P-02)
Void within session; same-till same-session refund; cross-session refund online by supervisor (H-10, FR-POS-050 corrected)	Discretionary discounts of any kind (ADR-14)
Park and resume; till session with float, drops, blind close and variance; Z-report, day sales and session report offline (H-11, I-03, O-02)	Till-to-till communication; shared parked sales; a shop server (doc 20 Appendix A)
Operator PIN, idle lock, cashier attribution; drawer-open-without-sale audited (N-03, N-09, H-13)	–
Shop-management mode hosting M4 (GRN, expected drops), M5 (count, write-off draft, repack, transfer receipt) and reports; role-gated; park-first; audited (ADR-36)	Online-only shop functions (orders, cross-session refunds, approvals) – the web back office on the same terminal
Local database with till-owned tables and read-only caches; local counters (till-position; location on the primary till); outbox; sync worker per doc 32; snapshot application with apply_from at day-open; heartbeat and telemetry	Update installation itself (management agent, doc 31)
Peripheral layer: scanner (2D), receipt printer (raster), drawer, scale, customer display, label printer; vendor adapters; simulator (H-07)	UnifiedPOS-certified drivers (vocabulary only)
Training mode on a sandbox namespace (T-04)	–
Central side: bundle validation and re-validation with the shared engine, till session and cash records, sync health and fleet views, cross-session refund handler	–

2. Requirements coverage

Identifier	Requirement (abridged)	Design section
FR-POS-010	Sale by scan, code, search; weighed goods	3.2, 3.6, 6.2, 6.3
FR-POS-020	Price resolution through the shared engine; MRP and control-price cap; picker; rules	3.2, 6.2, 6.4
FR-POS-030	Refund rules: same-till same-session at cashier level; cross-session online with supervisor	4.1, 6.7, 6.8
FR-POS-040	Negative stock permitted with explicit acknowledgement; exception event	3.2, 6.5
FR-POS-050 (corrected)	Online-only actions are blocked offline, not queued	3.8, 9.2
FR-POS-060	Park and resume	3.3, 4.1
FR-POS-070	Cashier attribution on every action; PIN; idle lock	3.5, 4.3
FR-POS-080	Receipt printing in the shop language with English fallback; raster	3.4
FR-POS-090	Offline set: sale, receipt, deduction, lookup, park, same-session refund, GRN capture, counts	3.8, 9.2
FR-POS-100	Shop-management mode on the same device	3.7, 4.4
FR-POS-110	Training/sandbox mode	3.9
FR-PAY-010	Tender types recorded, none processed	3.3
FR-PAY-030/040/050	Float, cash drops, blind close, variance reporting	3.5, 4.2
FR-REP-030	Z-report, day sales, session report offline	3.5
H-01, H-04	Two days offline; one writer, no merge	3.1, 3.10

Identifier	Requirement (abridged)	Design section
H-05, H-06, H-07, H-08	Cash only offline; ~1.5 tills; peripherals; dedicated terminals	3.6, 3.11
H-12, Q-07	Volume envelope; power resilience	9.1, 9.3
ADR-24, doc 31	Version reporting; self-test; floor behaviour	3.10, 6.12
Multi-MRP study	AUTO_LOWEST default; picker; 2D scan	3.2, 3.6

3. Domain model

3.1 Two databases, one logical model

Aspect	Design
Till-owned (written on the device, uploaded)	document (RCT and shop-mode types), document_line with pricing fields, doc_receipt_tender, document_state_history, stock_movement (sale deductions), audit_event, till_session, cash_movement, parked_sale, counters, outbox, sync_state. Same columns and types as doc 18 (doc 17 §9).
Central-owned (received read-only)	snapshot tables: sku, conversion, barcode, image (thumb keys), tag, tax_rate, batch, lot, price_list_line, discount_rule, control_price, mrp_policy, customer, operator, permission, count_task, expected_drop, expected_transfer, inventory_policy, config, message_catalogue, plus snapshot_version.
Central mirror	Schema pos at central: doc_receipt, doc_receipt_tender, till_session, cash_movement, device_health; everything else lands in the kernel base and the owning modules' tables through ingestion.
Encryption and durability	SQLite via Room with SQLCipher; key in the Android keystore; WAL; synchronous=FULL on issuance transactions; safe shutdown on battery threshold (Section 9.3).

3.2 Receipt

Aspect	Design
Document	RCT from the till-position series; header per doc 18 with business_date from the session; lines carry batch_id, unit_price, mrp_applied, control_price_applied, cap_reason, discount_rule_id, discount_amount, tax_rate, tax_amount, line_total (doc 18 document_line); header carries bill_rule_id, bill_discount, rounding, engine_version, snapshot_version.
Line creation	Scan → scan layer parses → local lookup (barcode, gtin+lot, INTERNAL code, or search) → LineInput → PriceResolver.resolveLine against the PricingSnapshotIndex built for the current snapshot version → policy: AUTO_LOWEST silently; BARCODE_RESOLVED uses the scanned batch; PICKER shows candidates in the ItemCard zone when the gap exceeds the threshold → line added; basket re-resolved for rules on every change (doc 23 §3.5).
Negative stock	If the resolved lot's cached quantity would go below zero, the line needs pos.negative_stock.acknowledge (one tap, reason NONE); the receipt line is flagged; an audit REVIEW event is recorded locally and uploaded (E-09).
Sell-through	An INACTIVE SKU with sell_through in the assortment sells with a badge; without it the lookup refuses with the reason (doc 22 DR-3).
Immutability	Issuance is the doc 18 §5.5 protocol on the device: counter increment and document insert in one local transaction; content_hash; nothing about an issued receipt changes; void and refund are new documents linked REVERSES / CREDITS.

3.3 Tenders, rounding and parked sales

Aspect	Design
Tenders	doc_receipt_tender(receipt, seq, kind CASH CARD_REF ACCOUNT VOUCHER (later) SUBSIDY (later), amount, reference, customer_account_id, change_given). Mixed tenders allowed; cash rounding applies to the cash portion of the total per the engine (G-10); change computed on cash. CARD_REF records the standalone terminal's approval reference keyed by the cashier; blocked offline (H-05).
Account tender	Against the cached customer balance and limit; offline permitted up to customers.offline_account_cap per receipt and the cached available balance; the till marks the tender offline = true; M7 reconciles on ingestion (doc 20 DR-3).
Parked sales	parked_sale(id, lines json, parked_at, operator) local only, per till; resumed by the same or another operator on the same till; expire at session close (a parked sale is not a document and has no number).

3.4 Receipt rendering and reprint

Aspect	Design
Content	Entity legal name and VAT number, shop name and address, receipt number (display form), business date and time, operator, lines (name in shop language with English fallback tag, qty × unit, MRP note when capped, rule name when reduced), subtotal, reductions, VAT summary by rate, rounding, total, tenders and change, gazette note when a control price bound a line, training watermark in sandbox; trilingual fixed strings from the message catalogue.
Rendering	Composed as a bitmap by the platform text stack with bundled Noto fonts and printed as raster through the printer adapter (P-05); the same renderer produces the Z-report, day sales and session report.

Aspect	Design
Reprint	Permitted for own-session receipts by the cashier and any receipt by the supervisor; every reprint is an audit event with the receipt number; the reprint carries a REPRINT banner.

3.5 Session and cash

Aspect	Design
Session	<code>till_session(id, position, operator, business_date, opened_at, float, closed_at, counted_cash, expected_cash, variance, closed_by)</code> ; one open session per position; the session's business date is the shop's current business date at open (doc 19 §9 day-close).
Cash movements	<code>cash_movement(session, kind FLOAT DROP PAYOUT? (no: not in scope) CLOSE_COUNT, amount, at, operator, witness?)</code> ; drops require a reason and print a slip; drawer open without a sale is an audited event (N-09).
Blind close	The cashier keys the counted cash without seeing the expected figure; the system computes <code>expected = float + cash tenders – change – drops</code> and the variance; the supervisor sees both; the session closes; the Z-report prints (I-03).
Day-close	When every open session at the location has closed (or the configured cut-off passes), the till advances the business date locally and activates pending <code>apply_from</code> rows; central's <code>DayCloseTrigger</code> does the same from the uploaded session events (doc 19 §9, doc 32 §5.2).

3.6 Scan layer and peripherals

Aspect	Design
Scan layer	Symbology detection; GS1 element-string and Digital Link parsing; AI 01 (GTIN), 10 (lot), 17 (expiry), 3103 (weight); dual-marked packs de-duplicated (single beep); 20–29 in-store labels decoded to SKU code + weight or + batch (doc 22 DR-6); output <code>ScanResult(symbology, gtin code, lot?, expiry?, weight?)</code> .
Peripheral layer	One interface per device class, UnifiedPOS vocabulary: <code>Scanner</code> , <code>PosPrinter</code> , <code>CashDrawer</code> , <code>Scale</code> , <code>LineDisplay</code> , <code>LabelPrinter</code> ; vendor adapters behind them (two or three vendors); a <code>Simulator</code> implementation for development, tests and training; health check per device at session open reported in telemetry (doc 14 §3).
Weighed goods	Scale read on demand (tare handled by the scale) or keyed weight with <code>pos.weight.key</code> ; the line carries the weight in KG and the per-kg price; weigh-station labels where a label printer exists.

3.7 Shop-management mode

Aspect	Design
Entry and exit	From the bottom bar; requires a role holding any shop-mode permission; the current sale must be parked first; entry and exit are audited (<code>MODE_SWITCH_SHOP_MGMT</code>); idle lock 3 minutes.
Hosted screens	GRN capture and confirm (doc 24A §8), count (doc 25A §8), write-off draft, repack, transfer receipt, expiry list, reprint, sync and device status, local reports. Each renders the owning module's specification on the shell components; the data comes from the snapshot rows those modules contribute.
Primary-till rule	Documents in the LOCATION series (GRN confirm, count submit, write-off submit, repack, transfer receipt) issue only on the primary till; a non-primary till captures DRAFTs and hands them over (the draft appears on the primary till after sync, or the web completes it). The tile shows which this device is.
Online-only from the terminal	Orders, statements, cross-session refunds, write-off submission and approvals open the web back office in the terminal's browser (kiosk allow-listed URL) with the operator's back-office credentials.

3.8 Offline and online sets (H-02 / H-03 as behaviour)

Works offline	Blocked offline (refused with reason, never queued)
Sale, receipt, stock deduction, lookup and search, rules and ceilings, park and resume, same-session void and refund, GRN capture and confirm (primary), count, write-off draft, repack (primary), transfer receipt (primary), account tender within cap, Z-report and session reports, reprint, PIN login, mode switch, training	Card reference confirmation of card, wallet or QR tenders; cross-session or cross-till refunds; write-off submission and any approval; new orders; master-data changes; anything MFA-gated; payment-receipt posting

3.9 Training mode

Aspect	Design
Namespace	A separate local database file and a separate numbering namespace (TRN - prefix, visibly non-live); the same snapshot; simulated peripherals or the real printer with a watermark; uploads to the sandbox entity at central so trainers can review (ADR-35).

3.10 Sync, counters and device state

Aspect	Design
Outbox	Every issued document bundle, movement set, audit event and session fact is appended with a dense device_seq in the same local transaction; the upload worker sends batches per doc 32 §3; acknowledged rows are retained 7 days then purged (S8).
Snapshots	Delta or full per doc 32 §5; staged, verified, swapped in one local transaction between sales; apply_from rows held pending until day-open; the PricingSnapshotIndex is rebuilt on swap.
Counters	counter(series_id, next_number) persisted with fsync inside the issuance transaction; the till-position series always; the location series only when this device is the primary till (flag from the snapshot); counter transfer per doc 32 §8.
Instructions	From acks and heartbeats: revoke (wipe caches, lock, preserve outbox), floor notice (banner; after grace sync refused, selling continues), re-request-from-seq (resend retained rows), series holder change (adopt or release the location counter).
Clock	Device clock plus monotonic uptime; offset from heartbeats; events carry occurred_local and the offset; sequence orders, never time.

3.11 Multi-till shops

Aspect	Design
Independence	Each till syncs alone and deducts against its own lot cache; oversell across tills is accepted (doc 25 §3.2); parked sales are per till; a session is per position; the primary till holds the location series. A shop-hub role is the phase-2 path if this bites (doc 20 Appendix A).

4. State machines

4.1 Sale and receipt

From	To	Trigger / command	Guard	Permission	Audit / offline
–	IN_PROGRESS	StartSale / add first line	session OPEN; operator logged in	pos receipt.issue	– / offline
IN_PROGRESS	PARKED	ParkSale	≥1 line	pos receipt.issue	SALE_PARKED / offline
PARKED	IN_PROGRESS	ResumeSale	same till; session OPEN	pos receipt.issue	SALE_RESUMED / offline
IN_PROGRESS	TENDERING	Tender	basket resolved; total > 0	pos receipt.issue	–
TENDERING	ISSUED	CompleteSale	tenders cover total (cash rounding applied); card ref only online; account within cap; issuance protocol (till-position series); movements; audit; outbox	pos receipt.issue	RECEIPT_ISSUED / offline
TENDERING	IN_PROGRESS	CancelTender	–	pos receipt.issue	–
IN_PROGRESS / PARKED	ABANDONED	AbandonSale(reason) or session close	reason when lines exist	pos receipt.issue	SALE_ABANDONED (REVIEW when lines > 0)
ISSUED	VOIDED	VoidReceipt(reason)	same session; not yet refunded; reason	pos receipt.void	RECEIPT_VOIDED (REVERSES -linked void document) / offline
ISSUED	PARTIALLY_REFUNDED / REFUNDED	RefundLines(receipt, lines, reason)	same till and session at cashier level; else supervisor online; qty ≤ unrefunded; reason	pos.refund.same_session / pos.refund.cross_session	REFUND_ISSUED (CREDITS-linked refund receipt) / offline for same session

4.2 Till session

From	To	Trigger / command	Guard	Permission	Audit / offline
–	OPEN	OpenSession(float)	no open session on the position; operator PIN; float ≥ 0; business date from day state	pos.session.manage	SESSION_OPENED / offline
OPEN	OPEN	CashDrop(amount, reason)	amount ≤ expected cash; slip printed	pos.session.manage	CASH_DROPPED / offline
OPEN	OPEN	OpenDrawerNoSale(reason)	reason	pos.drawer.open	DRAWER_OPEN_NO_SALE (REVIEW) / offline
OPEN	CLOSED	CloseSession(counted)	no sale in progress (parked sales abandoned with REVIEW); blind count keyed; variance computed; Z-report	pos.session.manage	SESSION_CLOSED (variance beyond threshold → REVIEW) / offline

4.3 Operator

From	To	Trigger / command	Guard	Permission	Audit / offline
–	LOGGED_IN	PinLogin	operator in cached snapshot; PIN matches Argon2id hash; not locked out	–	OPERATOR_LOGIN / offline
LOGGED_IN	LOCKED	Idle timeout or Lock	2–3 min sales, 3 min shop mode	–	–
LOCKED	LOGGED_IN	PinLogin	same operator; or another operator (switch is audited)	–	OPERATOR_SWITCH when different

From	To	Trigger / command	Guard	Permission	Audit / offline
any	LOCKED_OUT	5 failed PINs	–	–	AUTHENTICATION_FAILED × 5 → ACCOUNT_LOCKED_OUT (15 min)

4.4 Mode

From	To	Trigger / command	Guard	Permission	Audit / offline
SALES	SHOP_MGMT	SwitchToShopMode	sale parked or none; operator holds a shop-mode permission	any shop.* / inv.* offline permission	MODE_SWITCH_SHOP_MGMT / offline
SHOP_MGMT	SALES	ReturnToSales	no unsaved shop document (saved as draft first)	–	MODE_SWITCH_SALES / offline

4.5 Device (local view)

From	To	Trigger / command	Guard	Permission	Audit / offline
ENROLLED	ACTIVE	First snapshot applied; position assigned	–	system	–
ACTIVE	STALE	Snapshot older than staleness age	warning banner; selling continues	system	SNAPSHOT_STALE (REVIEW)
ACTIVE	BELOW_FLOOR	Floor notice past grace	sync refused; selling continues	system	VERSION_BELOW_FLOOR
any	REVOKED	Signed revoke instruction	cache wiped; screen locked; outbox preserved	system	DEVICE_REVOKED

5. Contract

5.1 Till-local commands (no REST; the till's own application layer)

Command	Guards	Effect	Uploaded as
OpenSession(float) / CashDrop / OpenDrawerNoSale / CloseSession(counted)	Section 4.2	session and cash rows; Z-report	till_session.opened/closed.v1; cash.dropped.v1; drawer.opened_no_sale.v1
StartSale / AddLine(scan code weight) / ChangeQty / RemoveLine / PickBatch / AcknowledgeNegativeStock	session OPEN; lookup success; engine result; policy	basket state (not a document)	– (audit for acknowledge and pick)
ParkSale / ResumeSale / AbandonSale	Section 4.1	parked_sale rows	audit events
Tender(kind, amount, reference account) / CancelTender / CompleteSale	coverage; online for CARD_REF; cap for ACCOUNT	receipt document, tenders, movements, audit, outbox	receipt.issued.v1 (bundle with movements)
VoidReceipt / RefundLines(same session)	Section 4.1	linked documents	receipt.voided.v1 / receipt.refunded.v1 (bundles)
ReprintDocument	own session or supervisor	print; audit	audit
PinLogin / Lock / SwitchOperator	Section 4.3	operator session	audit
SwitchToShopMode / ReturnToSales	Section 4.4	mode	mode.switched.v1
Shop-mode delegates: CaptureGrn / ConfirmGrn, StartCount / SubmitCount, RequestWriteOff (draft), ExecuteRepack, ReceiveTransfer	Owning modules' rules (docs 24A, 25A); primary-till rule	documents in the LOCATION series	the owning modules' bundles
SyncNow / ApplySnapshot / Heartbeat (worker)	doc 32	outbox, caches, state	–
EnterTraining / ExitTraining	operator holds pos.training	namespace switch	sandbox uploads

5.2 Central commands and queries

Item	Description	Permission / scope
RefundCrossSession(receipt, lines, reason)	Supervisor on the web (or the till online): issues a refund receipt CREDITS-linked to the original from the supervisor's till-position series if at a till, else the location's ENTITY-scoped refund series (DR-1)	pos.refund.cross_session (online)
SetOfflineAccountCap / SetIdleLocks / SetReceiptTemplate	Configuration register items	pos.config (entity)
GetSyncHealth(device location entity) / ListStuckDevices	Pending counts, last ack, snapshot version, floor state, anomalies	OWN; FEDERATION_VIEW; helpdesk
GetTillSessions(location, date) / GetVariances(scope, period)	Sessions with blind-close variances	OWN, FEDERATION_VIEW; M8
GetReceipt(number) / SearchReceipts(location, date, amount)	For cross-session refunds and disputes	OWN
DeviceHealth(device)	Peripheral self-test, battery, storage, version, clock offset	OWN; DevOps; doc 31

5.3 Events (uploaded by the till or raised at central)

Event	Payload	Consumers
receipt.issued/voided/refunded.v1 (bundles)	Document, lines with pricing fields and batch, tenders, movements, state row, engine and snapshot versions	M5 (deductions), M7 (account tenders), M8, kernel re-validation
till_session.opened/closed.v1; cash.dropped.v1; drawer.opened_no_sale.v1	Session facts, variance	M8 (variance), exception queues, day-close trigger
mrp_cap.applied.v1 / mrp_picker.used.v1 / negative_stock.acknowledged.v1	Receipt line, reason, batch	M3 reviews, exception queues, M8
mode.switched.v1; operator.switched.v1	Device, operator, mode	Audit
till.heartbeat.v1; device.health.v1; snapshot.applied.v1	Versions, pending, battery, peripherals, offset	Fleet console (doc 31), freshness (M8), sync health

Event	Payload	Consumers
sync.local_anomaly.v1	What the till could not apply (a snapshot row it cannot parse)	DevOps alert

6. Key flows

Flow	Sequence	Failure paths
6.1 Open the day	Cashier PIN → OpenSession with float → peripheral self-test (printer test line, scanner beep, drawer) → pending apply_from rows activated if the business date advanced → sales screen	A failed printer self-test warns and allows selling with receipts queued for reprint; a failed scanner falls back to code entry
6.2 The three-tap sale	Scan (line appears with thumbnail, price, MRP note) → scan → [Cash] → total rounded, tendered amount keypad or exact → change shown large → receipt prints → drawer kicks; under a second from last scan to print	Unknown barcode → code or name search; no list line → 'not sellable' with reason
6.3 Weighed item	Scan the 20–29 weigh label (SKU + weight) or select the item and read the scale → line at per-kg price; ceiling checked per kg	Scale unreachable → keyed weight with permission and an audit event
6.4 Two batches, picker policy (doc 13 scenario 4 variant)	Scan → engine returns needsPick with two candidates (MRP 980 and 1,080, gap above threshold) → ItemCard shows both with pack images → cashier taps the one matching the pack → line with cap_reason MRP_PICKED and mrp_picker.used.v1	Under AUTO_LOWEST the same scan produces the 980 line silently
6.5 Negative stock	Scan an item whose cached lot is at 0 → inline prompt 'records show none in stock; sell anyway?' → one tap (permission) → line flagged; REVIEW event	Without the permission the line is refused and a supervisor PIN can approve
6.6 Park and resume	Customer forgot their wallet → Park → next customer served → Resume from the parked list	Parked sales left at session close are abandoned with a REVIEW
6.7 Same-session refund	Customer returns an item bought an hour ago → cashier selects the receipt from own session → lines and qty → reason → refund receipt (CREDITS link), cash returned, SALE_REVERSAL movement	A receipt from a previous session → 'supervisor online' path (6.8)
6.8 Cross-session refund	Supervisor logs in online (till or web) → searches the receipt by number → refund lines with reason → refund document; audit with both actors	Offline → refused with reason; the customer is asked to return when online or to the MPCs office
6.9 Blind close	End of shift → CloseSession → cashier keys counted cash without the expected figure → variance computed → Z-report prints → supervisor sees variance; beyond threshold → REVIEW	Sale in progress must be completed or abandoned first
6.10 Power cut mid-sale	Battery keeps the terminal up; at the low threshold the app completes or abandons the sale, flushes the database and shuts down cleanly; on restart the last transaction is either fully present or absent (spike 1)	A hard cut below threshold: WAL recovery on start; the property is still all-or-nothing per issuance transaction
6.11 Two days offline, then reconnection	Trading continues; a lorry arrives and the GRN is confirmed on the primary till; a count is done; the snapshot ages (staleness banner at 3 days) → connectivity returns → upload worker sends batches in sequence, acks arrive, retained rows marked → snapshot delta downloaded, staged, swapped between sales → central ingests, M5 deducts and records oversells, M4 processes the GRN	Ack lost → resend same batch id (DUPLICATE outcomes); central restored from backup → re-request-from-seq served from retained rows
6.12 A gazette lands while trading	Urgent change-log row → next heartbeat downloads the delta → swap between sales → the next scan of the affected item caps at the new ceiling with a gazette note on the receipt	Till offline: the cap applies on reconnection; central re-validation flags any sale above the ceiling made in between with the snapshot version evidence (S4)
6.13 Version floor	Floor raised; grace banner for 14 days → the agent installs outside trading hours (doc 31) → if not installed after grace: sync refused, selling continues, helpdesk queue	–
6.14 Suspension	Device reported stolen → revoke instruction on next contact → caches wiped, screen locked, outbox preserved for administrator recovery	–

7. Configuration

Item	Default	Scope	Who may change
pos.idle_lock_sales / shop_mode (minutes)	2 / 3	Entity; location override	gov.user.manage
pos.pin_length, lockout attempts / minutes	4–6; 5 / 15	Entity	gov.user.manage
pos.cash_rounding	NEAREST_RUPEE (engine)	Federation	–

Item	Default	Scope	Who may change
pos.card_reference_required	true	Entity	pos.config
customers.offline_account_cap (per receipt)	Rs 5,000	Entity	pos.config
pos.session_variance_review_threshold	Rs 500 or 1%	Entity	pos.config
pos.parked_sale_max_age (minutes)	120	Entity	pos.config
pos.reprint_limit_per_receipt	3	Entity	pos.config
pos.receipt_template	default (fields fixed; header and footer text per entity)	Entity	pos.config
pos.snapshot_staleness_days	3	Federation	–
pos.battery_safe_shutdown_pct	8	Federation	–
pos.quiet_hour (24-hour shops, for updates)	02:00–04:00	Location	pvt.location.*
pos.training_allowed_roles	cashier, supervisor, shop-in-charge	Entity	gov.role.manage

8. Screens as paper

Screen	Paper mirrored	Roles	Offline	Notes
PIN and session	Attendance book	Operator	Yes	Float entry; self-test results; blind close; cash drop with slip
Sales	Till slip	Cashier	Yes	Zones per doc 30 §4.3; ItemCard hosts picker, negative-stock prompt, sell-through badge, weight entry
Park / resume	–	Cashier	Yes	Parked list with age and first line
Refund (same session)	Refund slip	Cashier	Yes	Own-session receipts; line picker; reason codes
Refund (cross session)	Refund slip	Supervisor	No	Receipt search; both actors recorded
Weigh and price	Scale ticket	Cashier	Yes	Scale read or keyed weight
Reports	Z-report, day sheet	Supervisor	Yes	Print via raster renderer; last 7 sessions
Shop-management home	Shop diary	Shop-in-charge	Yes	Tiles: receive goods, count, write-off, repack, transfer, expiry, reprint, sync status; primary-till badge
Sync and device status	–	Supervisor	Yes	Pending uploads, last ack, snapshot version and age, floor state, peripherals, battery, version
Training	–	All permitted	Yes	Loud badge and watermark
Web on the terminal	–	Shop-in-charge	No	Kiosk allow-listed back office for online-only functions

9. Non-functional design

Aspect	Design
9.1 Performance	Scan to line under 300 ms on the reference terminal (lookup on indexed local tables, engine under 5 ms); receipt render and print start under 1 s; snapshot swap under 2 s for a 6,000-SKU assortment; the basket index rebuilt on swap, not per scan (<i>doc 23 §9.1</i>).
9.2 Offline	Two days: about 400 receipts per till per day at the planning basis, 2–4 KB each with movements; the local database stays under 200 MB including caches and thumbnails; 7-day retention of acknowledged rows about 10 MB (<i>doc 32 §10</i>).
9.3 Power and durability	Internal battery (<i>doc 14 §3.1</i>); safe shutdown at the battery threshold after completing or abandoning the current sale; WAL with synchronous=FULL on issuance; the spike-1 harness injects power loss at every step of issuance and asserts all-or-nothing and dense counters.
9.4 Security	Kiosk lock-task; device-owner managed; keystore-held database key; Argon2id PIN hashes; operator snapshot signed; no card data anywhere; USB debugging and unknown sources disabled by policy; revoke wipes caches and locks (<i>doc 31 §8</i>).
9.5 Privacy	Customer cache holds name, phone, balance and limit only; receipts do not print the phone number; the audit log stores identifiers (<i>ADR-27</i>).
9.6 Usability	Three taps to sell; 56 dp targets; keypad-first; no modal chains; reason codes instead of confirmations; loud training mode; two cashier rounds before freeze (<i>doc 30 §2.3, spike 7</i>).
9.7 Observability	Telemetry on every heartbeat; crash reports; sync anomalies; a local diagnostics screen for the helpdesk within its boundaries (no connectivity troubleshooting).

10. Test strategy

Layer	What is tested
Unit (JVM)	Every guard in Section 4; tender coverage and rounding; blind-close arithmetic; scan-layer parsing (GS1 fixtures, dual marking, 20–29 labels); engine parity with the golden baskets (<i>doc 23A</i>); outbox sequencing; snapshot staging and swap; apply_from activation
Instrumentation (emulator and reference terminal)	Issuance transaction with power-loss injection (spike 1); database encryption; kiosk behaviour; peripheral adapters against the simulator and real devices (spike 3); raster receipts in three languages (spike 2)
Sync conformance	The thirteen cases of <i>doc 32 §11</i> against the central simulator; two-day backlog drain; central restore re-request
Scenario	Flows 6.1–6.14; <i>doc 13</i> scenarios 2 and 4; two tills overselling; GRN, count, write-off draft, repack and transfer bundles accepted by central
Soak	48-hour disconnected trading on the reference terminal with periodic power cuts; storage growth and battery behaviour measured
Usability	Two rounds with real cashiers (spike 7): task completion, tap counts, error rates in Sinhala, Tamil and English
Central	Bundle re-validation flags but never rejects; session and variance records; cross-session refund handler; sync health queries

11. Decision requests

#	Point	Proposed	Decider
DR-1	Cross-session refund series	Issued from the supervisor's till-position series when done at a till; from a location-scoped refund series when done on the web	Federation accounts; kernel
DR-2	Non-primary tills and LOCATION-series documents	Drafts only; primary till or web confirms	Till track; Federation admins
DR-3	Card reference mandatory for CARD_REF tenders	Yes; online only	Federation accounts
DR-4	Offline account tender cap default	Rs 5,000 per receipt within cached available balance	Federation with MPCs treasurers
DR-5	Parked sale maximum age and session-close behaviour	120 minutes; abandoned at close with REVIEW	Federation
DR-6	Receipt legal content	As Section 3.4; confirm with the tax advisor when their review arrives (24B advisory)	Programme

#	Point	Proposed	Decider
DR-7	Customer display in phase 1	Where the terminal has a second screen; line, total, change	Federation (doc 30 DR-4)
DR-8	Keyed weight without a scale	Permitted with pos.weight.key and an audit event	Federation
DR-9	Helpdesk kiosk-exit procedure	Supervisor PIN plus a time-boxed code from the helpdesk; audited	DevOps, helpdesk lead (doc 38)

12. Traceability

Section	Identifiers
3.1, 3.10, 3.11	ADR-10, ADR-34; H-01, H-04, H-06; NFR-OFF-010/030/040; doc 32; doc 18 §5.5
3.2–3.4	FR-POS-010 to 040, 080; ADR-14, ADR-31; G-10; H-09; P-05; N-05; multi-MRP study; doc 23 §3.5
3.3	FR-PAY-010; I-01, I-02; ADR-19; H-05; doc 20 DR-3
3.5	FR-PAY-030 to 050; FR-REP-030; I-03, I-04; O-02; doc 19 §9
3.6	FR-POS-010; H-07; B-11; doc 14 §3; doc 22 DR-6
3.7–3.9	FR-POS-100, 110; ADR-36; H-02, H-03; T-04; ADR-35
4	Doc 03 §13; N-03, N-09; FR-POS-030, 060, 070
6	Doc 13 scenarios 2 and 4; doc 14 §8 spikes 1–7; doc 31
9	NFR-PRF, NFR-OFF, NFR-SEC-020, NFR-USE; Q-07

End of document. Draft for design review; the contract freezes when M7 and M8 acknowledge it and the till track signs the bundle specifications of docs 24A and 25A, targeted end of Stage 2 week 6. Implementation guide: 26A.